

13.1 Farm-to-Table 4 All

Farm-to-Table 4 All is a nonprofit 501(c)(3) organization providing healthy food choices in low income neighborhoods. Dena, Eric, and Cullen are its founders and current directors and believe that nutritious food choices are part of a healthy lifestyle. They have three restaurant locations: North, East, and South. At all three locations and every day, those restaurants serve meals including a small appetizer and a dessert.

They are considering opening a fourth location (West) and need to construct a personnel budget comprised of cooks and servers. The fourth location is projected to serve 300 meals per day. Here is a list of information they have compiled from the other locations

- **Worker Productivity:** One cook can prepare 140 meals per day. One server can handle 105 customers per day.
- **Work Time:** Each employee works five days a week.
- **Wage:** Cooks earn \$34 per hour and servers earn \$19 per hour.
- **Fringe Benefits:** The fringe benefits (i.e., retirement contribution, Social Security, and unemployment insurance) amount to 21.2% of the total salary cost. In addition, the cost for health insurance amounts to \$100 per employee per month.
- **Weekend Bonus:** Since the restaurant will be open every day, Farm-to-Table 4 All will need to pay a salary bonus of 25% for work during the weekend.
- **Vacation Time:** All employees have ten paid vacation days.

Provide the three founders with a detailed personnel budget to deliver 300 meals a day and given the above parameters. The expenses need to be calculated for a year, i.e., it will be an annual budget. The following components should be included in the budget (separated by cooks and servers):

1. **Workload calculations and personnel needed:** In a first step, calculate the meals served per year, which is straight forward. Next, calculate the work (in days per year) that can be delivered by the cooks and servers. This allows you to calculate the number of cooks and servers you need (round up).
2. **Base salary budget:** Calculate the hours per year that cooks and servers are working. This calculation is necessary since the wage is given in \$ per hour. Make sure to include the vacation days as well.
3. **Weekend salary budget:** Calculate how many employees you need on weekends and how much you have to pay them just for working on the weekend.
4. **Total budget:** In a last step, combine the numbers of the base and weekend salary and add the fringe benefits. This final step should give you the total personnel budget.

The three founders asked you to keep the budget flexible to assess the effects of

key parameters on the cost. Specifically, they are interested how the following situations affect the bottom line:

1. Increase in the hourly wage by 15% for both cooks and servers. This may be necessary to overcome the labor shortage.
2. Decrease in the number of meals served to 250 due to higher prices and consumers decreasing their restaurant visits.

The founders are also considering whether AI could assist Farm-to-Table 4 All with its budgeting and workforce planning. AI-based tools can potentially use historical information on customer demand, staffing, wages, and operating conditions to forecast future meal demand and identify alternative staffing scenarios. Assume that the organization is considering using an AI-based forecasting system to support preparation of the personnel budget for the West location. The system would not make the final budget decision; instead, it would provide forecasts and scenarios for management to evaluate.

Using an AI tool, develop three plausible scenarios for the number of meals served per day at the West location: a low-demand scenario, a baseline scenario, and a high-demand scenario. For each scenario:

1. Calculate the number of cooks and servers required.
2. Calculate the resulting annual personnel budget, including base salaries, weekend salary premiums, health insurance, and fringe benefits using the AI tool.
3. Compare the personnel costs across the three scenarios and calculate the personnel cost per meal.
4. Explain how errors in the AI demand forecast could affect the organization's annual budget. Consider separately the consequences of overestimating and underestimating demand.
5. Identify at least three pieces of historical data that would be useful for training or informing an AI-based demand forecasting model.
6. Evaluate the AI-generated scenarios critically. Identify any assumptions or recommendations produced by the AI that appear unreasonable, and explain how you would verify them before incorporating them into the organization's budget.

Finally, write a short recommendation to the founders explaining how AI could be used as a decision-support tool in the organization's budgeting process while maintaining human oversight and accountability for the final budget.